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|-----------------------------------------------------------------------------------|-----------------|-------------------------|-------|
|  | NAME            |                         |       |
|                                                                                   | REGISTRATION NO |                         | MARKS |
|                                                                                   | CLASS           | DDTS IS1 / IS2          |       |
|                                                                                   | CODE / SUBJECT  | DFSS0113 SECURITY AUDIT |       |
|                                                                                   | ASSESSMENT      | PRACTICAL TASK 2        |       |
|                                                                                   | LECTURER        |                         |       |

**CLO 1 : PERFORM APPROPRIATE TECHNIQUES IN AUDITING WEB AND NETWORK USING VARIOUS TOOLS (P4, PLO 3)**

**TOPIC : 3.2 PERFORM SECURITY TESTING USING APPROPRIATE TOOLS FOR AUDITING NETWORK**

**DURATION : 3 HOURS**

**INSTRUCTION: ANSWER ALL THE QUESTIONS**

The practice of reviewing and analyzing a computer network for security vulnerabilities and loopholes is known as network vulnerability assessment. Network administrators use it to assess a network's security architecture and defense against possible attacks and vulnerabilities.

Metasploitable2 is an intentionally vulnerable Linux distribution and is also a highly effective security training tool. It comes fully loaded with many vulnerable network services and includes several vulnerable web applications.

**Machine required:**

1. Kali Linux (Pen Testing Machine)
2. Metasploitable2 (Target)

**Question:**

By using **Nmap and Nmap Script Engine (NSE)**, perform the following task below:

1. Displays scanning target IP Address.
2. Make connectivity with target.
3. Displays the open ports and state their services.
4. Perform SMB enumeration to discover user accounts on the target system.
5. Perform enumeration of RPC-related services to discover port/protocol service.

**Note**

1. Your report must include:
  - Cover.
  - Table of contents
  - Screenshot and explain every process.

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# INTRODUCTION

Network security assessment is an essential process in identifying and mitigating vulnerabilities within a network system. It involves using various tools and techniques to detect open ports, misconfigurations, and weak points that attackers could exploit. In this task, Kali Linux is used as the penetration testing machine, and Metasploitable2 is used as the target vulnerable system. The Nmap and its Script Engine (NSE) are utilized to perform different types of scans and enumeration such as discovering active IP addresses, identifying open ports and services, enumerating users through SMB, and detecting RPC related services. This practical exercise helps students understand real-world network auditing and vulnerability detection.

# QUESTION

## 1. DISPLAYS SCANNING TARGET IP ADDRESS.

First, in kali linux I put command `ifconfig` to recognize my IP addresses.

```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.192.130 netmask 255.255.255.0 broadcast 192.168.192.255
    inet6 fe80::41e9:ae52:fd8e:245d prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:78:e1:cc txqueuelen 1000 (Ethernet)
    RX packets 52 bytes 5390 (5.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 542 bytes 34398 (33.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Then, after I got my IP. I put command `netdiscover -r <my IP kali>`  
This command is to scan network range to find target IP (Meta).

```
(kali@kali)-[~]
$ sudo netdiscover -r 192.168.192.130
```

After scanning, it will show like this it means the scan is succeed.

```
Currently scanning: Finished! | Screen View: Unique Hosts
4 Captured ARP Req/Rep packets, from 4 hosts. Total size: 240
+---+-----+-----+-----+-----+-----+
| IP | At MAC Address | Count | Len | MAC Vendor / Hostname |
+---+-----+-----+-----+-----+-----+
| 192.168.192.1 | 00:50:56:c0:00:08 | 1 | 60 | VMware, Inc. |
| 192.168.192.2 | 00:50:56:e6:b6:d1 | 1 | 60 | VMware, Inc. |
| 192.168.192.133 | 00:0c:29:6b:02:88 | 1 | 60 | VMware, Inc. |
| 192.168.192.254 | 00:50:56:f0:dc:6d | 1 | 60 | VMware, Inc. |
```

After that, I check the IP for metasploitable using the same command in kali. And the IP is the same on the scanner.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:6b:02:88
          inet addr:192.168.192.133  Bcast:192.168.192.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe6b:288/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:897 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:56087 (54.7 KB)  TX bytes:11238 (10.9 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:142 errors:0 dropped:0 overruns:0 frame:0
          TX packets:142 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:44105 (43.0 KB)  TX bytes:44105 (43.0 KB)
```

192.168.192.133

## 2. MAKE CONNECTIVITY WITH TARGET.

For this part is very easy, we just need to put `ping` command on 2 machine meta and kali. Then switch the IP between that machine. If the packets not have loss it means the connection is good.

This pinging from kali to meta

```
(kali㉿kali)-[~]  
$ ping 192.168.192.133  
PING 192.168.192.133 (192.168.192.133) 56(84) bytes of data.  
64 bytes from 192.168.192.133: icmp_seq=1 ttl=64 time=0.788 ms  
64 bytes from 192.168.192.133: icmp_seq=2 ttl=64 time=0.453 ms  
64 bytes from 192.168.192.133: icmp_seq=3 ttl=64 time=0.458 ms  
^C  
— 192.168.192.133 ping statistics —  
3 packets transmitted, 3 received, 0% packet loss, time 2036ms  
rtt min/avg/max/mdev = 0.453/0.566/0.788/0.156 ms
```

This pinging from meta to kali

```
msfadmin@metasploitable:~$ ping 192.168.192.130  
PING 192.168.192.130 (192.168.192.130) 56(84) bytes of data.  
64 bytes from 192.168.192.130: icmp_seq=1 ttl=64 time=0.377 ms  
64 bytes from 192.168.192.130: icmp_seq=2 ttl=64 time=0.336 ms  
  
--- 192.168.192.130 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 999ms  
rtt min/avg/max/mdev = 0.336/0.356/0.377/0.027 ms
```

The both machine have no packets loss it mean it is work.

### 3. DISPLAYS THE OPEN PORTS AND STATE THEIR SERVICES.

For this part, im using command `nmap -p- <IP META>`. This command is to scan all open ports and services on the target to find vulnerabilities.  
Many open port and service on this target.

```
(kali㉿kali)-[~]
$ nmap -p- 192.168.192.133
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-21 02:12 EDT
Nmap scan report for 192.168.192.133
Host is up (0.0010s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
36016/tcp open  unknown
37443/tcp open  unknown
47117/tcp open  unknown
50614/tcp open  unknown
MAC Address: 00:0C:29:6B:02:88 (VMware)
```



## 4. PERFORM SMB ENUMERATION TO DISCOVER USER ACCOUNTS ON THE TARGET SYSTEM.

This is the user account that have been discovered.

The command I use is `nmap --script smb-enum-users.nse -p445 192.168.192.133`.

It use to enumerate SMB user on the target.

I don't know why all of the account disabled but this is the way to discover user account in target system.

```
(kali㉿kali)-[~]
$ sudo nmap --script smb-enum-users.nse -p445 192.168.192.133

Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-21 02:22 EDT
Nmap scan report for 192.168.192.133
Host is up (0.00032s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 00:0C:29:6B:02:88 (VMware)

Host script results:
| smb-enum-users:
|   METASPLOITABLE\backup (RID: 1068)
|     Full name:  backup
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\bin (RID: 1004)
|     Full name:  bin
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\bind (RID: 1210)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\daemon (RID: 1002)
|     Full name:  daemon
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\dhcp (RID: 1202)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\distccd (RID: 1222)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\ftp (RID: 1214)
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\games (RID: 1010)
|     Full name:  games
|     Flags:      Account disabled, Normal user account
|   METASPLOITABLE\gnats (RID: 1082)
```



## 5. PERFORM ENUMERATION OF RPC-RELATED SERVICES TO DISCOVER PORT/PROTOCOL SERVICE.

For this part, im using `nmap --script rpcinfo 192.168.192.133`.  
This command use to identify RPC services and their associated ports.  
It have rpcbind,nfs,status.

```
(kali㉿kali)-[~]
$ nmap --script rpcinfo 192.168.192.133

Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-21 02:27 EDT
Nmap scan report for 192.168.192.133
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
| rpcinfo:
|   program version   port/proto  service
|   100000   2             111/tcp    rpcbind
|   100000   2             111/udp    rpcbind
|   100003   2,3,4        2049/tcp   nfs
|   100003   2,3,4        2049/udp   nfs
|   100005   1,2,3        36002/udp  mountd
|   100005   1,2,3        36016/tcp  mountd
|   100021   1,3,4        43776/udp  nlockmgr
|   100021   1,3,4        50614/tcp  nlockmgr
|   100024   1             47117/tcp  status
|_  100024   1             47280/udp  status
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:6B:02:88 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

# CONCLUSION

Through this practical task, we successfully used Nmap and its Script Engine (NSE) to audit a vulnerable network system. The exercise demonstrated how to identify an active host, test connectivity, discover open ports, and enumerate various services and user accounts. These findings provide valuable insights into how attackers might exploit vulnerable systems and help network administrators strengthen their network defenses. Overall, this activity improved understanding of vulnerability assessment, penetration testing, and the importance of regular network auditing to maintain system security.